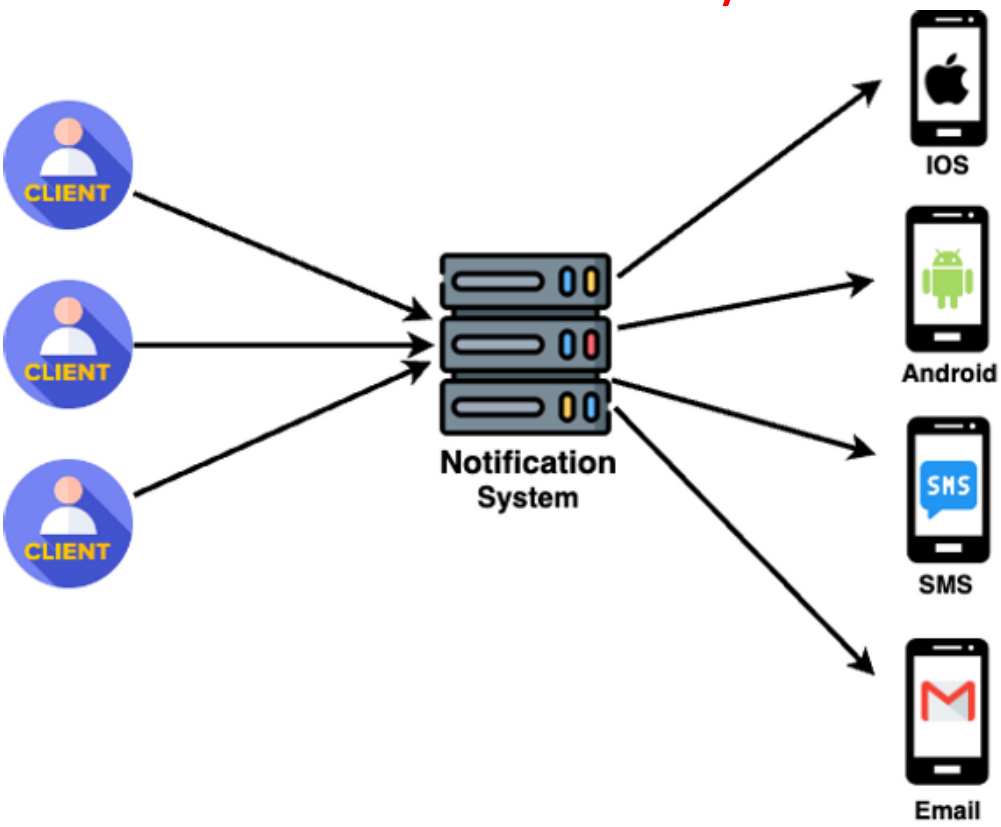
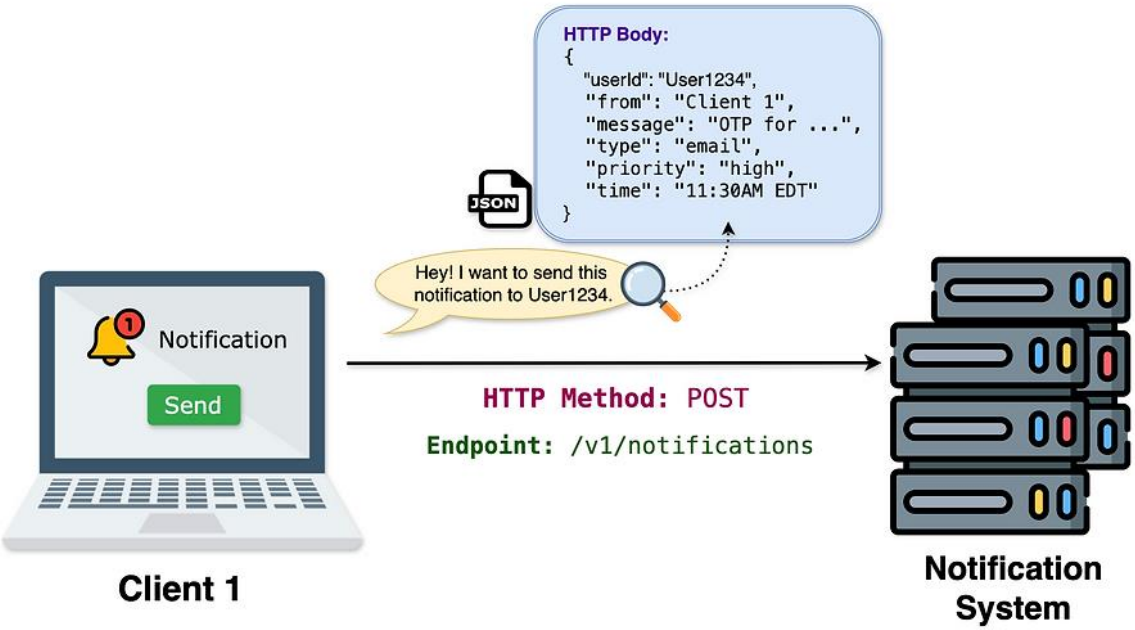
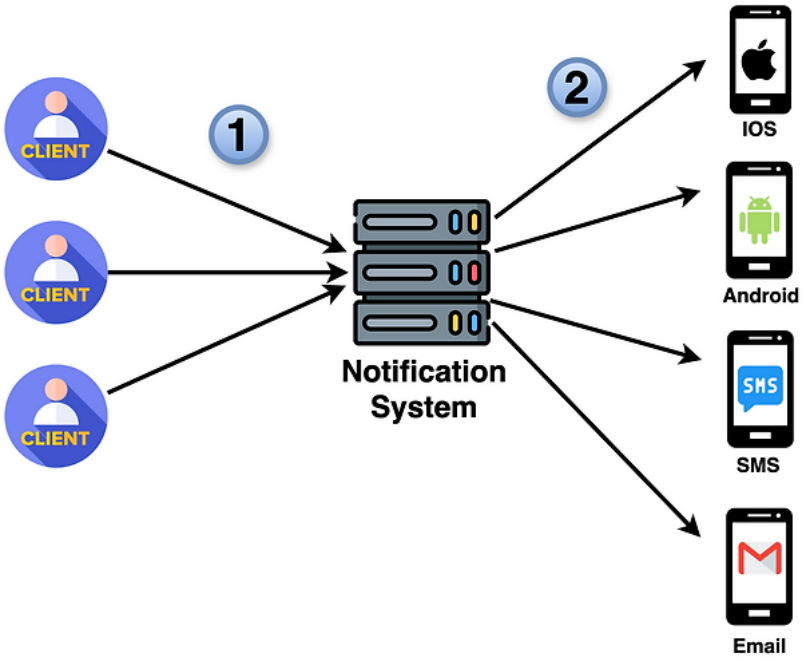
What is Notification System?



API Design- Sending a Notification

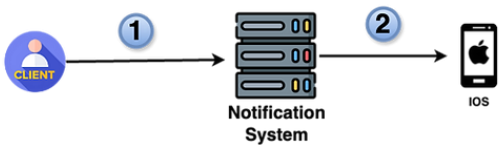


High Level Design- Basic Flow

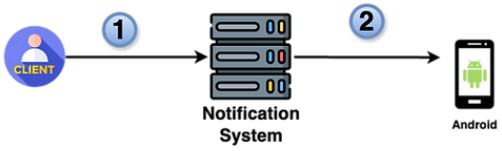


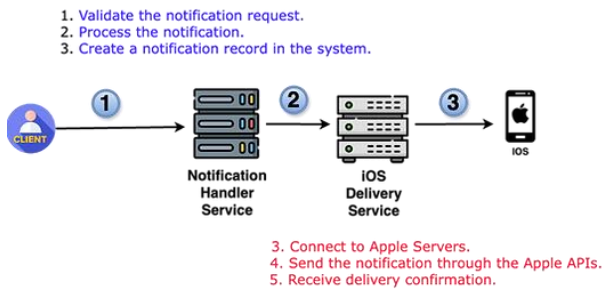
High Level Design- Problems with Basic High Level Design

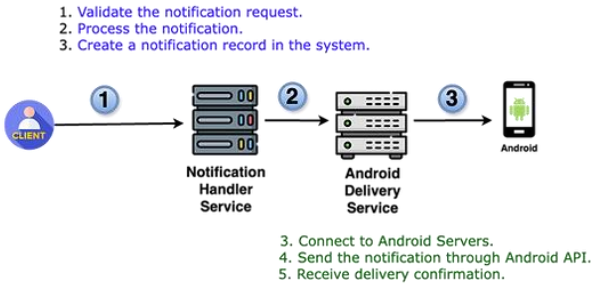
1. Validate the notification request.
2. Process the notification.
3. Create a notification record in the system.
4. Connect to Apple Servers.
5. Send the notification through the Apple APIs.
6. Receive delivery confirmation.

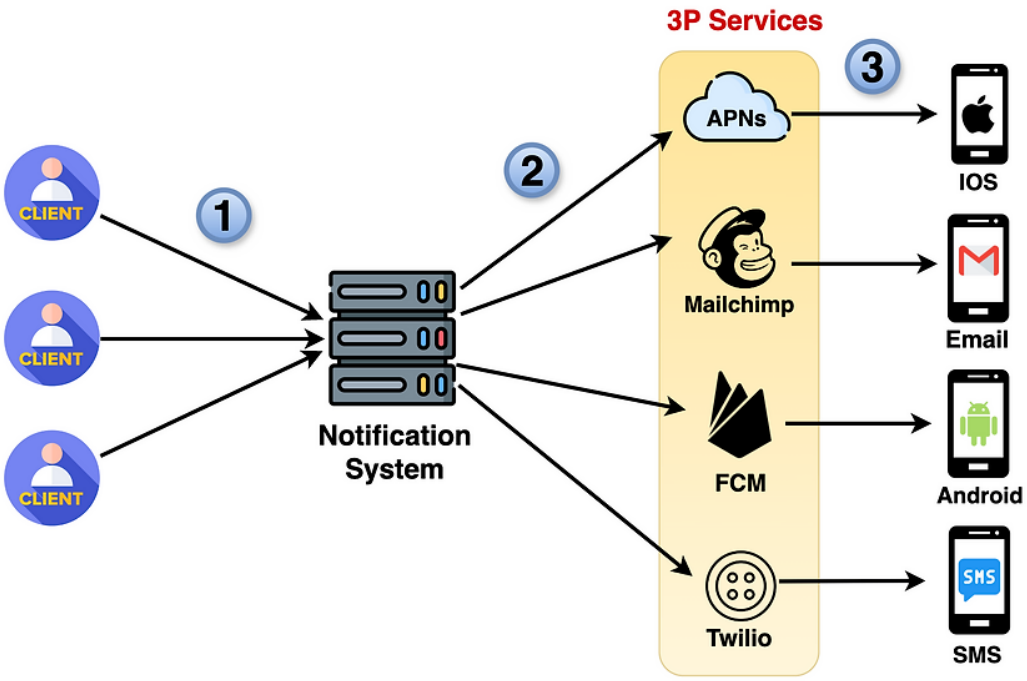


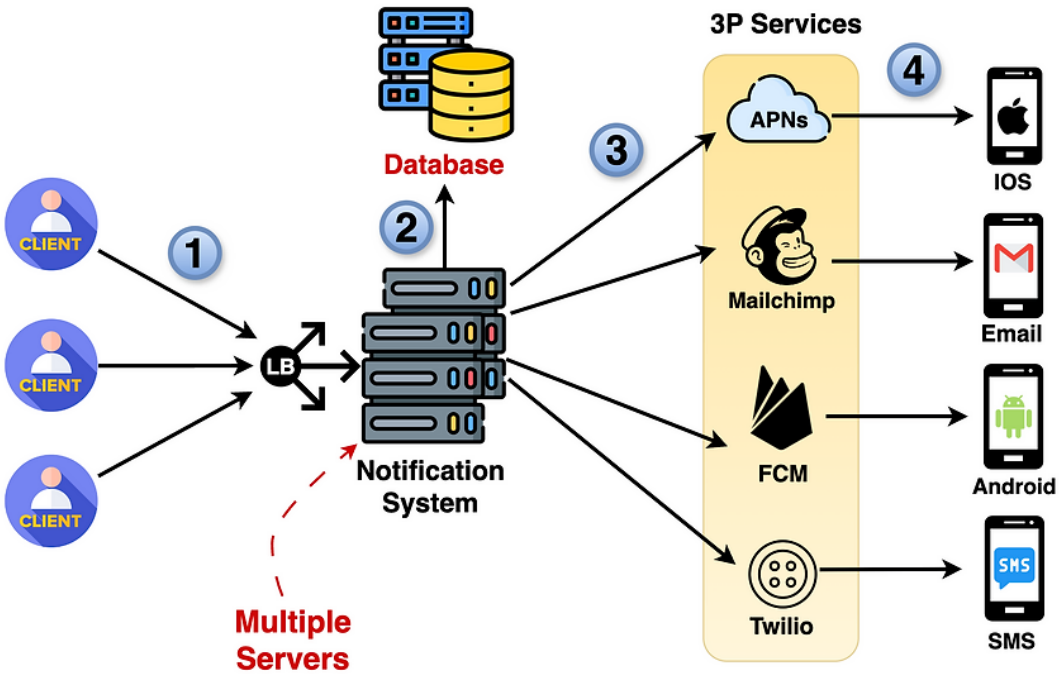
1. Validate the notification request.
2. Process the notification.
3. Create a notification record in the system.
4. Connect to Android Servers.
5. Send the notification through Android APIs.
6. Receive delivery confirmation.



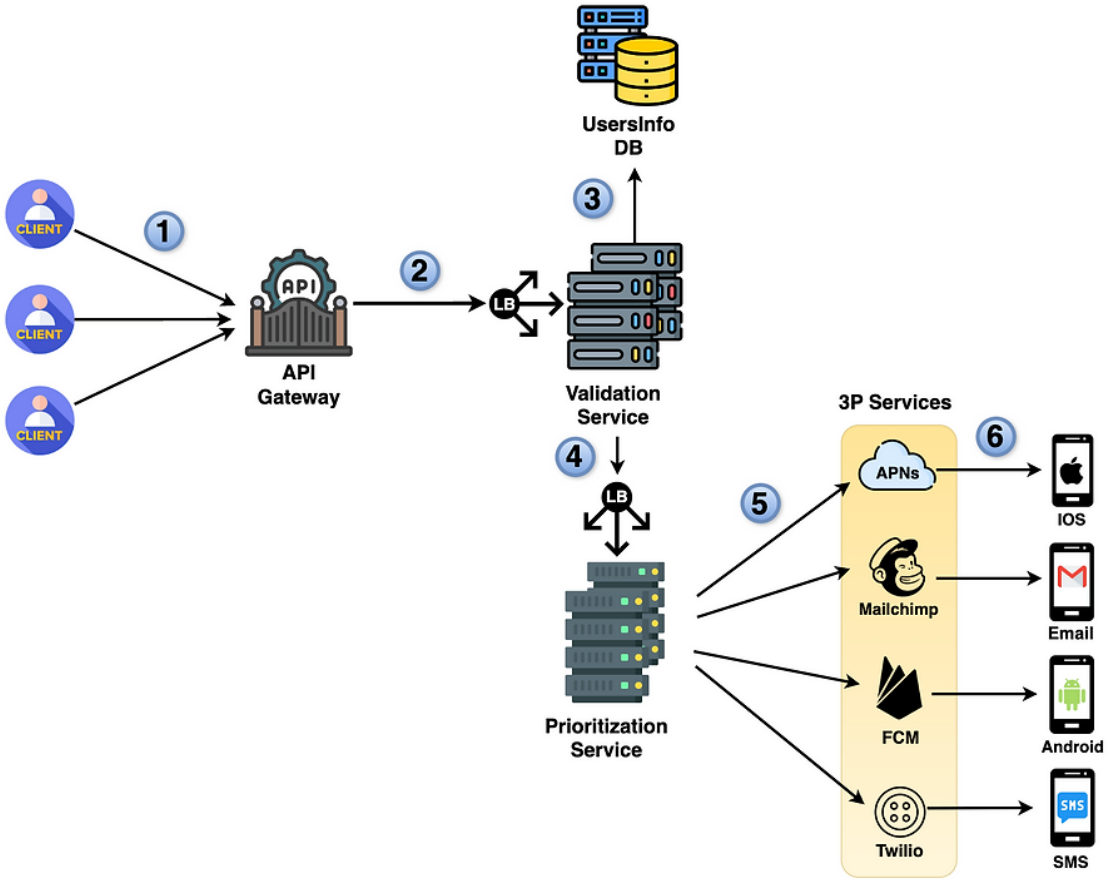

1. Validate the notification request.
2. Process the notification.
3. Create a notification record in the system.
CLIENT
1
2
3
Notification Handler Service
iOS Delivery Service
IOS
3. Connect to Apple Servers.
4. Send the notification through the Apple APIs.
5. Receive delivery confirmation.


1. Validate the notification request.
2. Process the notification.
3. Create a notification record in the system.
CLIENT
1
2
3
Notification Handler Service
Android Delivery Service
Android
3. Connect to Android Servers.
4. Send the notification through Android API.
5. Receive delivery confirmation.

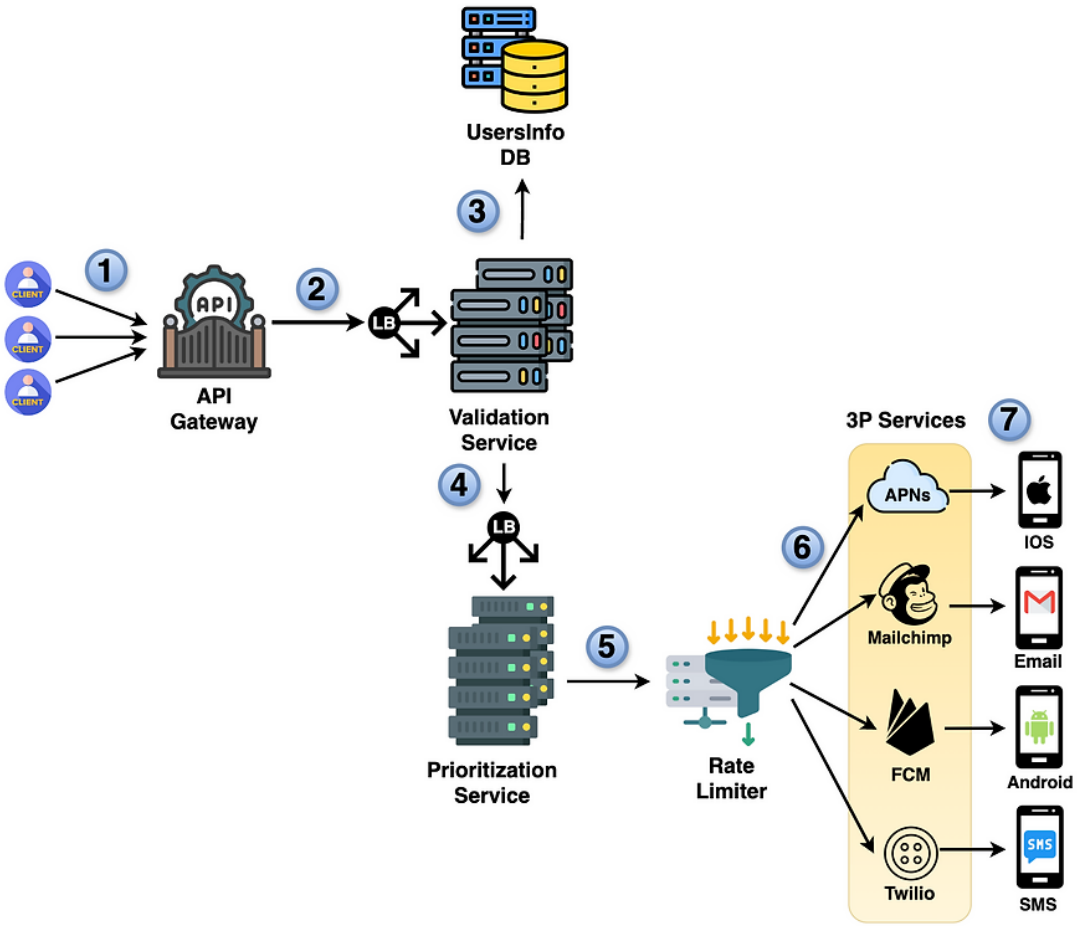

3P Services
CLIENT
CLIENT
CLIENT
1
2
3
Notification System
APNs
Mailchimp
FCM
Twilio
IOS
Email
Android
SMS



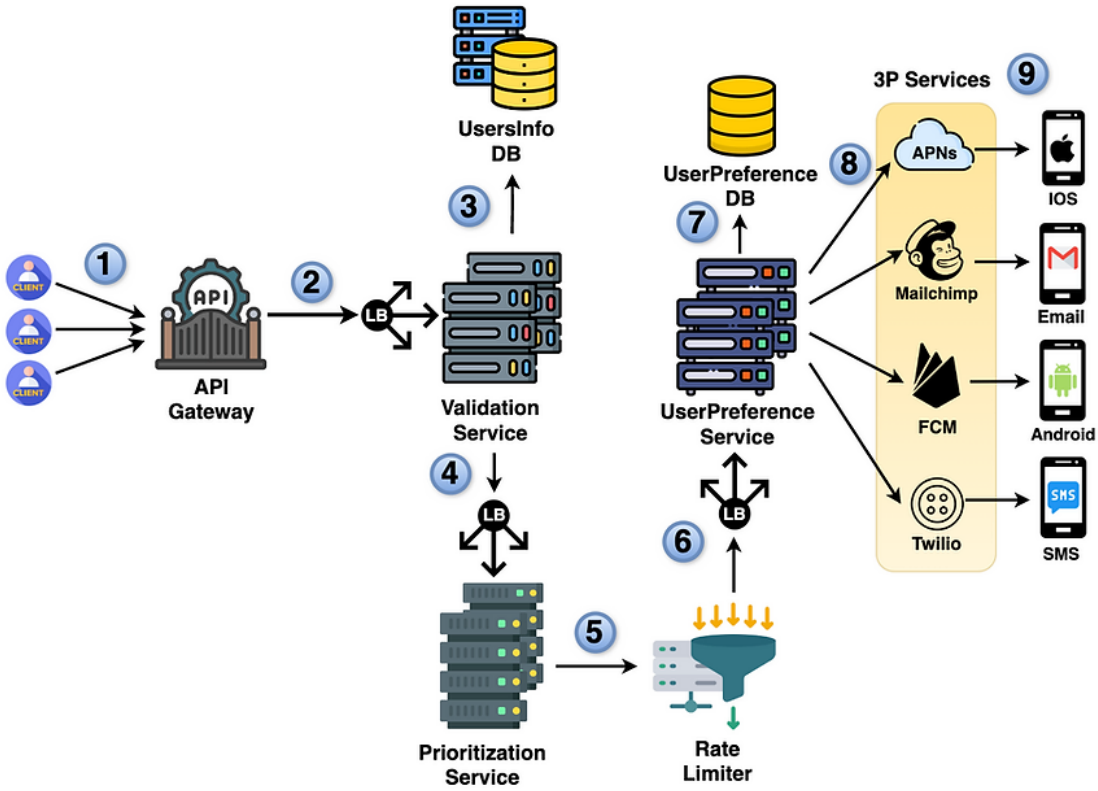
High Level Design-Validation and Prioritization



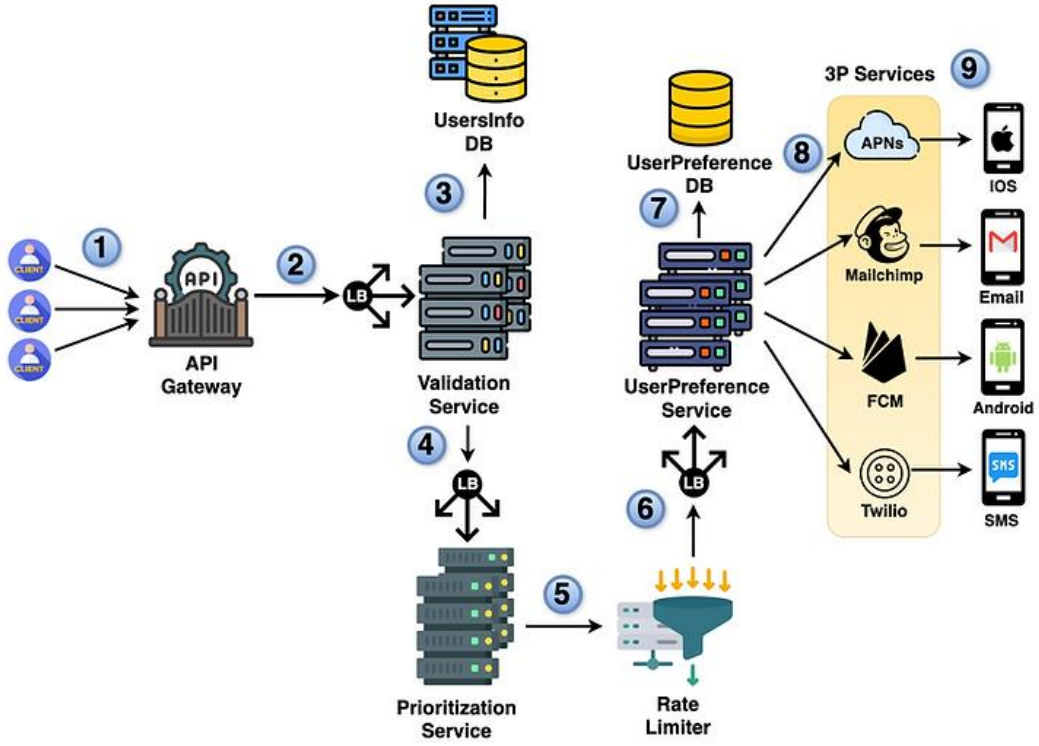
High Level Design-Rate Limiting Notifications

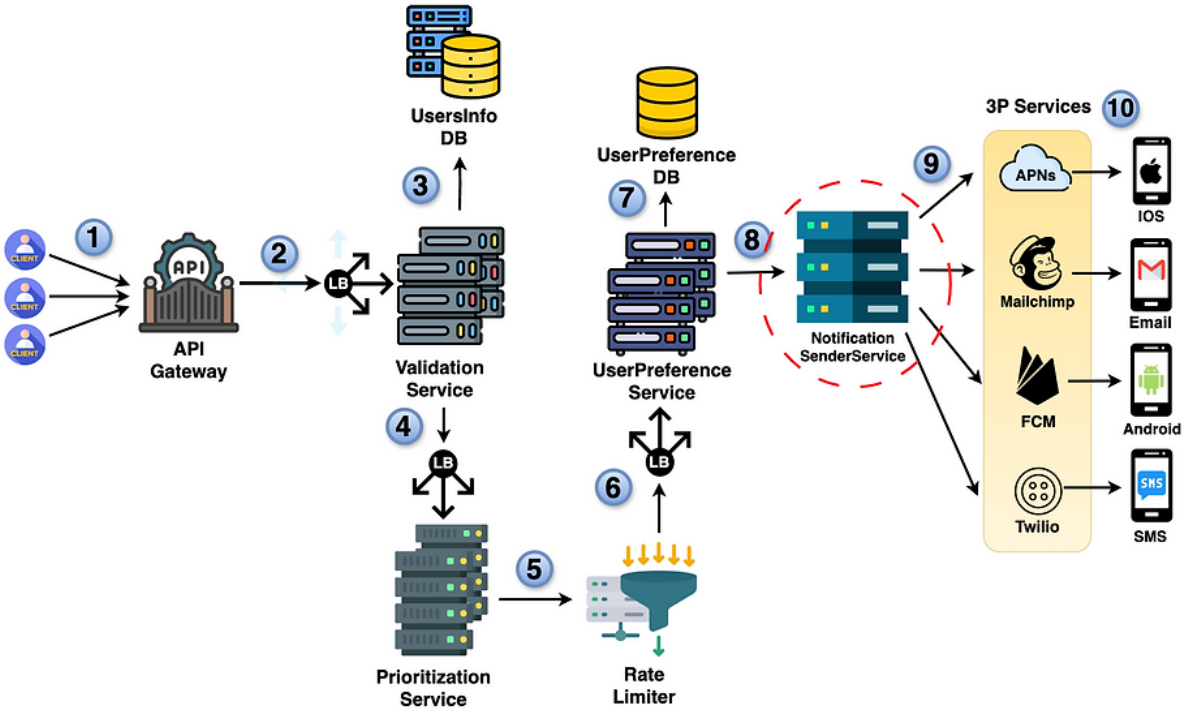


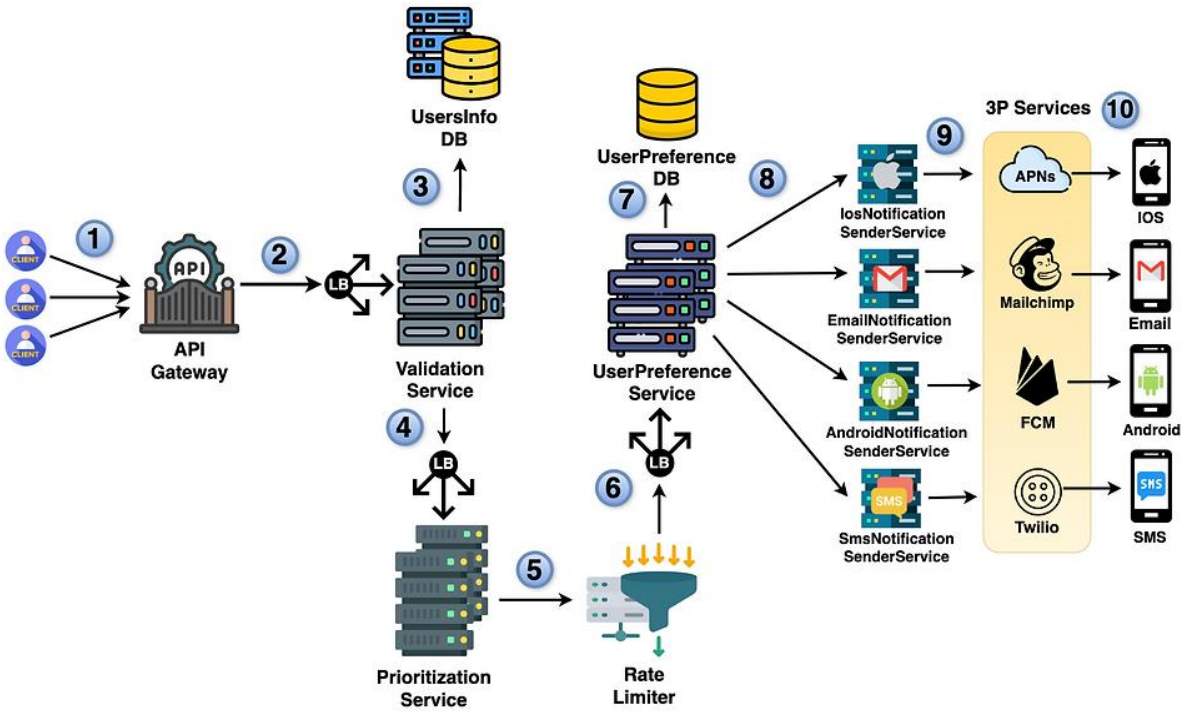
High Level Design-Adjusting to User Preferences



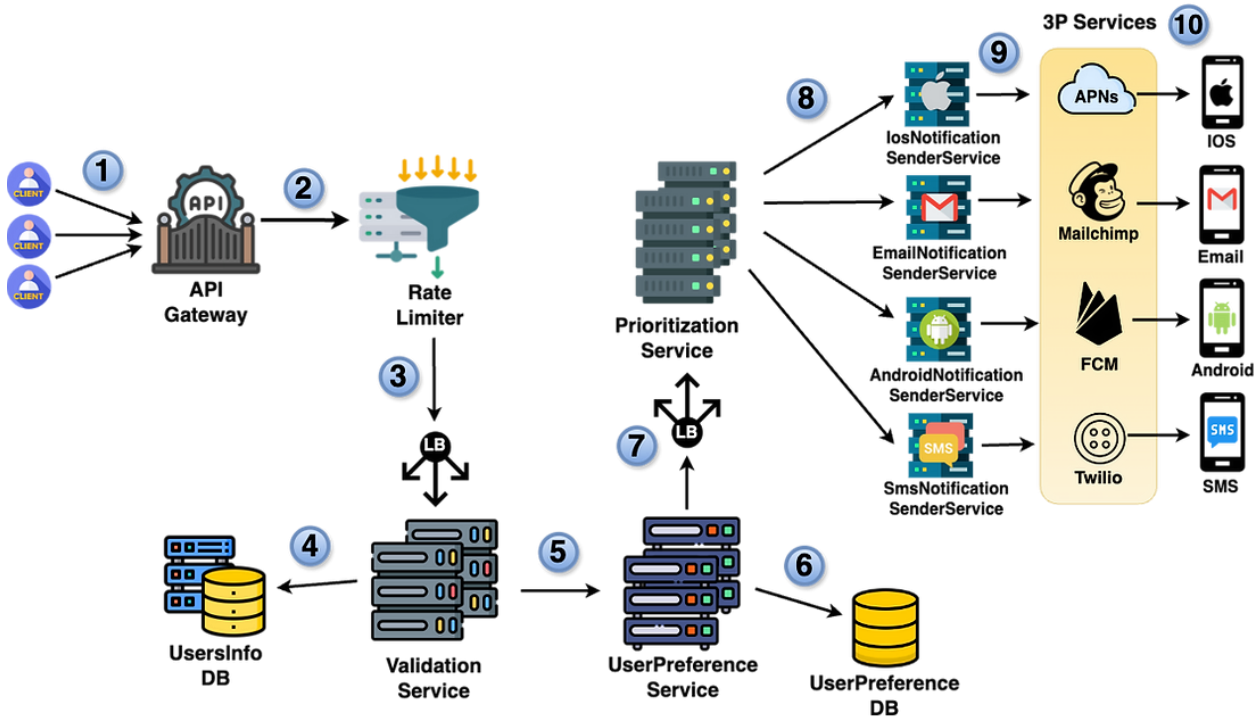
High Level Design-Handling different types of Notifications



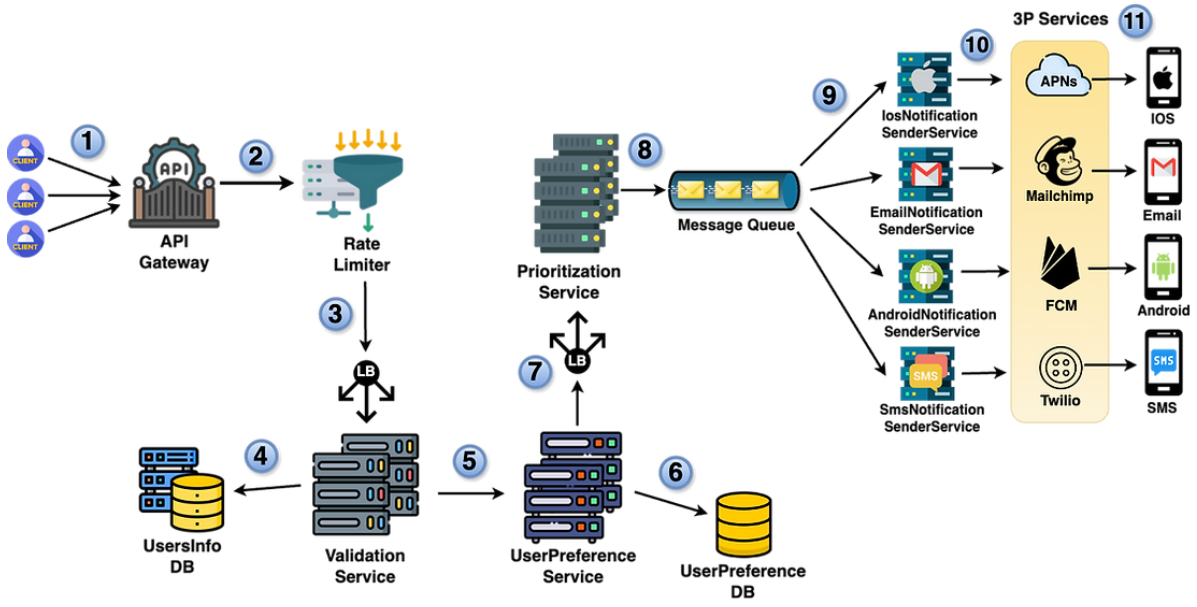




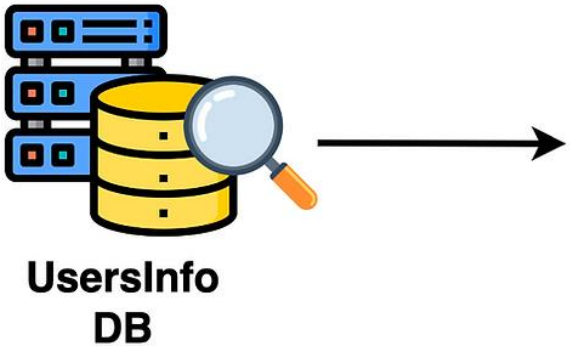
High Level Design-Reorganizing Services



High Level Design-Decoupling & Optimizing Design

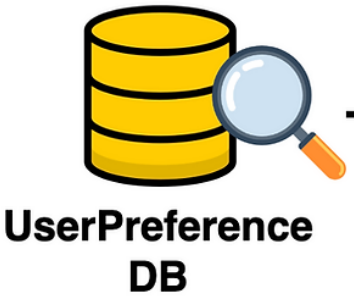


Deep Dive Insights- Database Modeling (User Info DB Schema)



Column Name	Data Type	Description
userId	VARCHAR	The unique identifier for each user (PK).
name	VARCHAR	The user's full name.
email	VARCHAR	The user's email address (unique).
phoneNumber	VARCHAR	The user's phone number (unique).
isActive	BOOLEAN	Indicates if the user's account is active.
createdAt	TIMESTAMP	When the user was created.
updatedAt	TIMESTAMP	When the user's information was last updated.

Deep Dive Insights- Database Modeling (User Preference DB Schema)




```
{
  "userId": "unique_user_id",
  "preferences": {
    "email": {
      "optIn": true,
      "notificationTypes": ["orderConfirmation", ..]
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    ...
  },
  "createdAt": "1720789150",
}
```
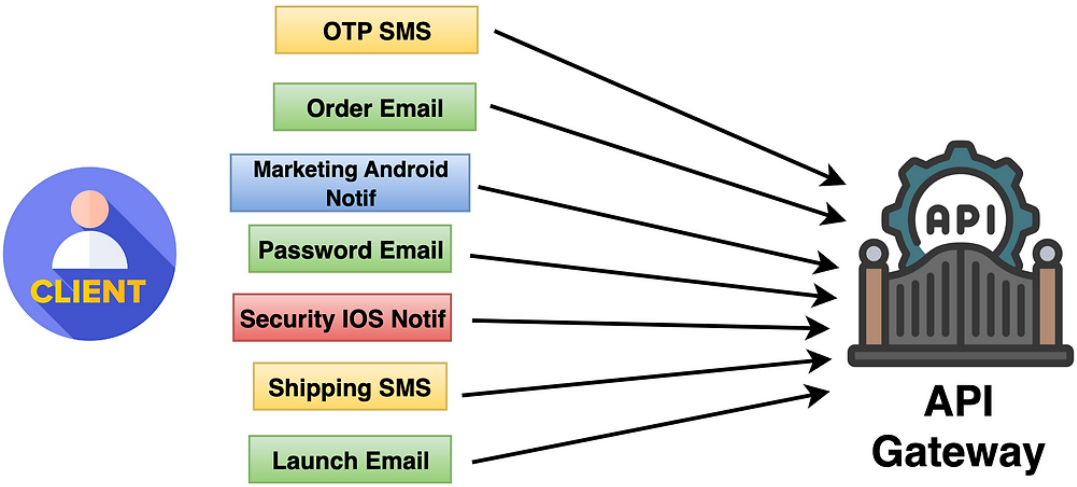
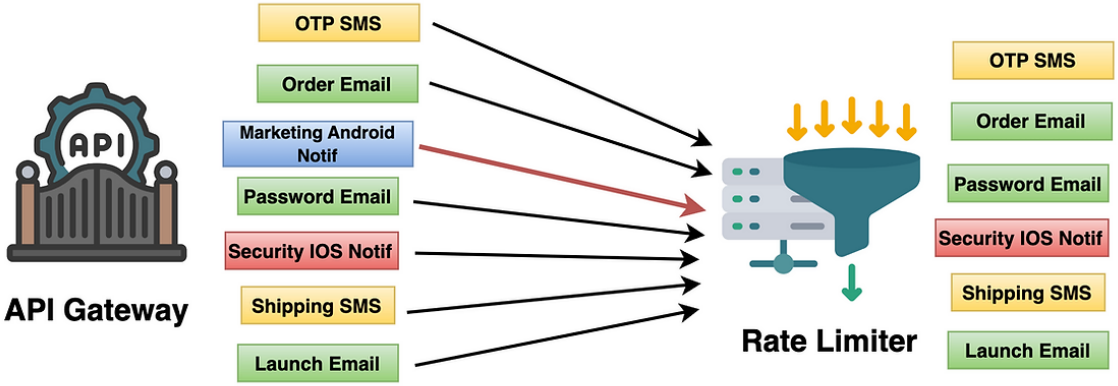

Deep Dive Insights- Notifications Overall Flow

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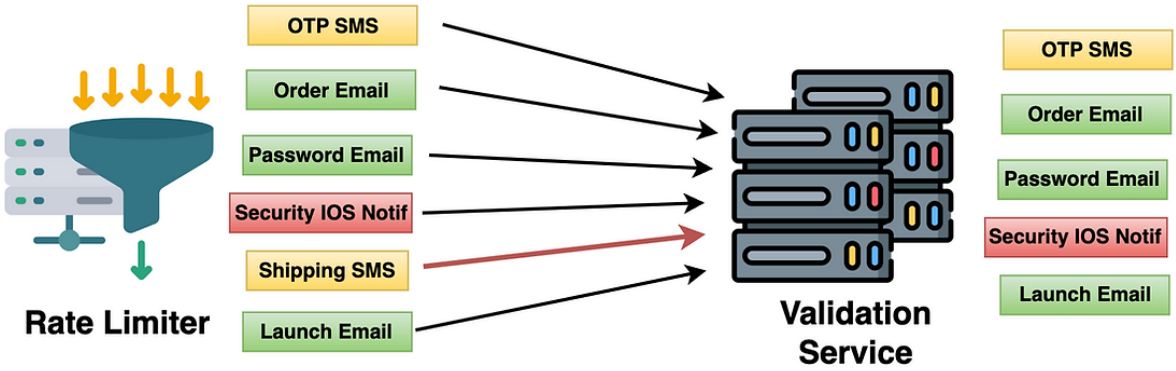
API Gateway
OTP SMS
Order Email
Marketing Android Notif
Password Email
Security IOS Notif
Shipping SMS
Launch Email
Rate Limiter
OTP SMS
Order Email
Password Email
Security IOS Notif
Shipping SMS
Launch Email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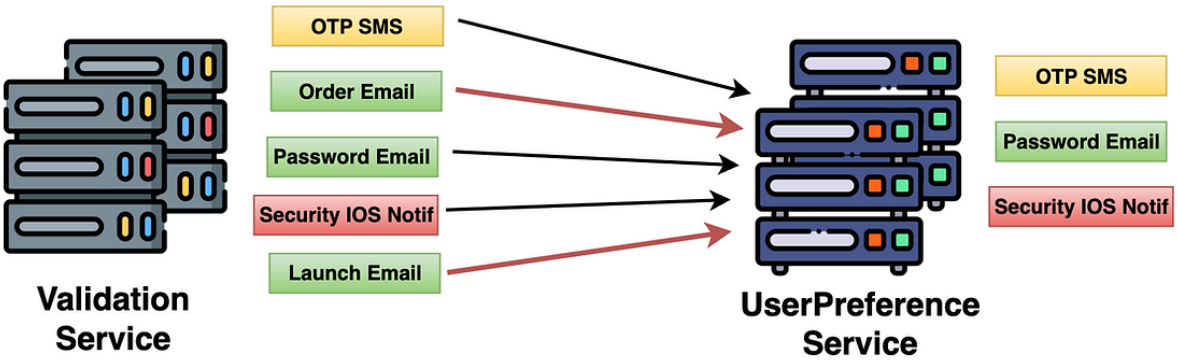
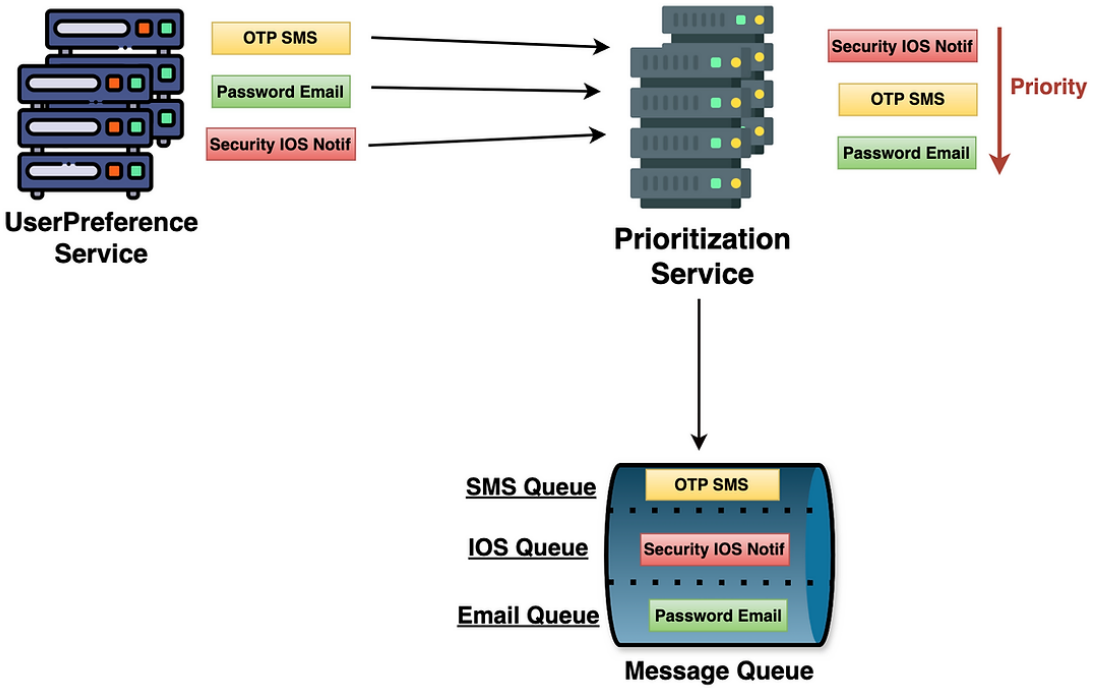
Password Email
Security IOS Notif
Shipping SMS
Launch Email
Validation Service
OTP SMS
Order Email
Password Email
Security IOS Notif
Launch Email


Validation Service
OTP SMS
Order Email
Password Email
Security IOS Notif
Launch Email
UserPreference Service
OTP SMS
Password Email
Security IOS Notif


OTP SMS
Password Email
Security IOS Notif
UserPreference Service
Prioritization Service
Security IOS Notif
OTP SMS
Password Email
Priority
SMS Queue
IOS Queue
Email Queue
OTP SMS
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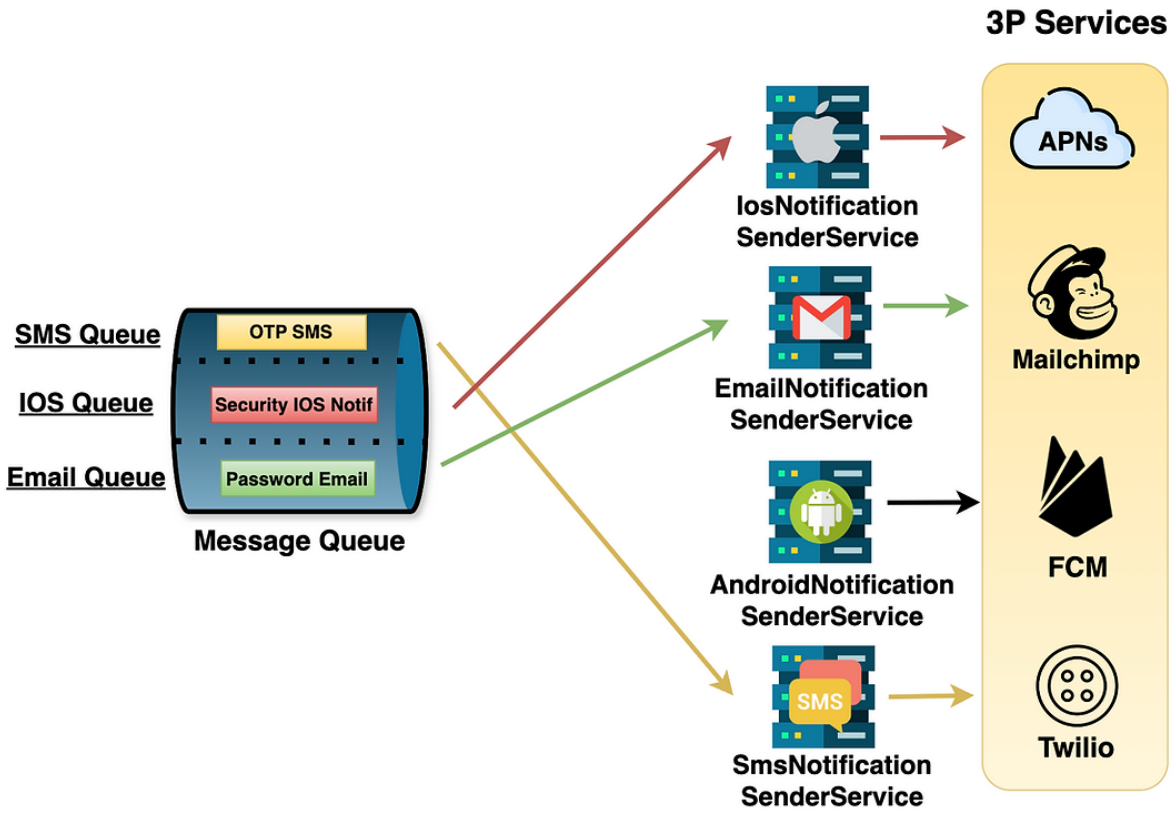
Password Email
Message Queue


3P Services
SMS Queue
IOS Queue
Email Queue
OTP SMS
Security IOS Notif
Password Email
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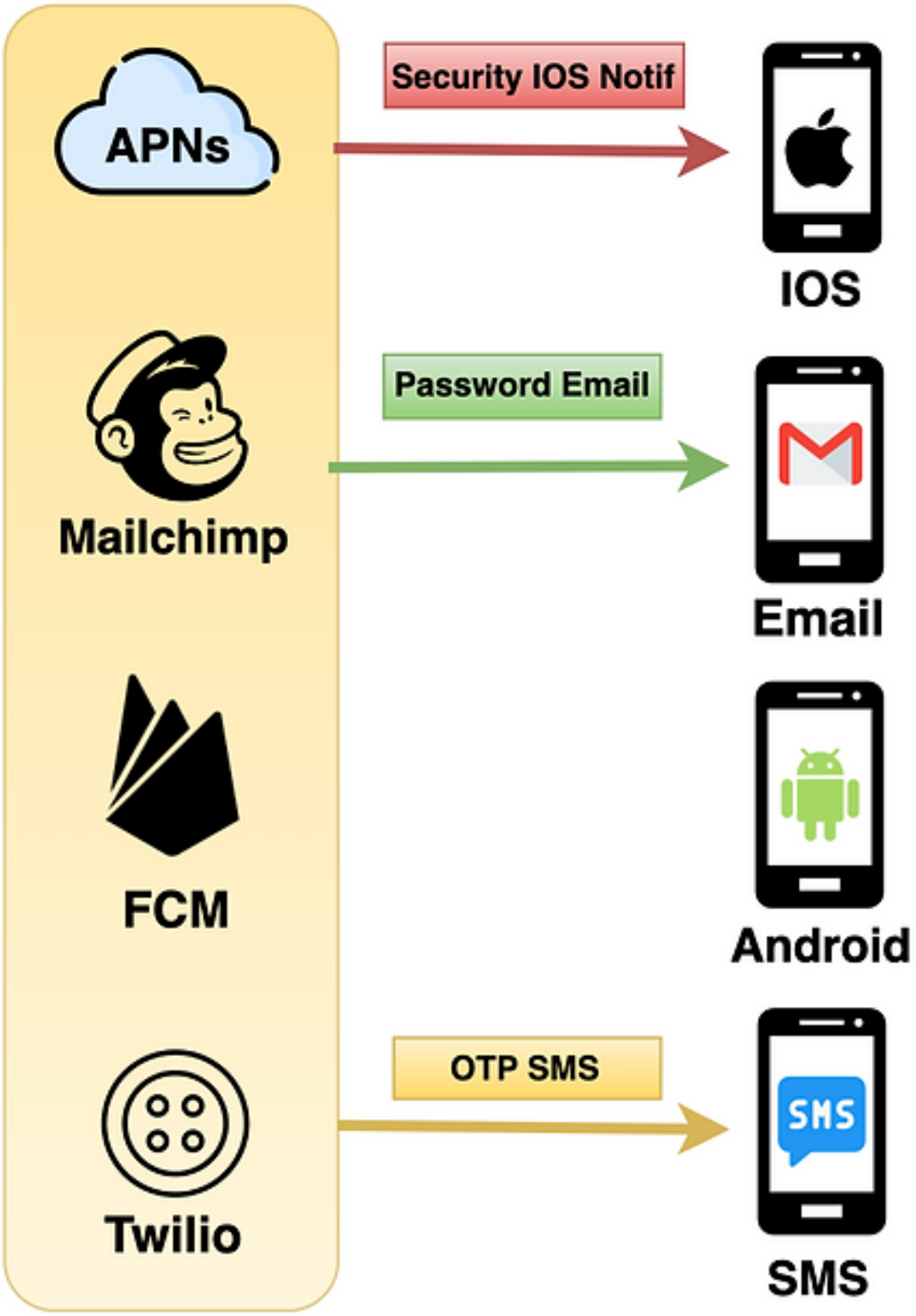
3P Services


APNs
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